HYPNEXIS'26 · AI-01 · PHASE 2

Kavach Phase 2: real phone calls and partly faked speech

Plan, method and live status. Numbers marked "pending" are filled in automatically from runs/phase2/results.json when each experiment finishes. No number in this document is typed by hand.

1. What Phase 2 asks

#	Requirement	Pass condition
R1	Report EER after 8 kHz resampling, two codecs and added noise	Worst case ≤ 2.5 × clean EER
R2	Build 100 partial fakes and locate the synthetic segments	Segment F1 at 0.5 s resolution (report it)
R3	Streaming mode during a call	Score updates every 500 ms
R4	Keep Phase 1 results	Phase 1 EERs within 2 points of before

2. Our approach

R1 · Channel robustness

Real calls pass through narrowband phone lines and lossy voice codecs, with background noise. We test each condition on its own, on 2,000 clips from 11 In-the-Wild speakers never seen in training:

Condition	How it is made	Why
Clean	Silence-trimmed, loudness-normalised 4 s clip	Reference EER
8 kHz resample	Down to 8 kHz and back to 16 kHz	Telephone bandwidth: everything above 4 kHz is lost
Codec 1: AMR-NB 12.2 kbps	Real encoder (ffmpeg libopencore_amrnb) at 8 kHz	The standard mobile-network voice codec (GSM/UMTS calls)
Codec 2: Opus 12 kbps VoIP	Real encoder (ffmpeg libopus, voip mode)	WhatsApp, Google Meet and most internet calls
Added noise	Street/room-like noise at 10 dB SNR	Calls from busy places
Random mix	Random chain of the above, plus speaker-to-mic replay	Hardest realistic case

Training change: the detector is re-trained with random channel augmentation (8 kHz, one of four codecs: AMR, Opus, GSM, MP3, noise at 5–25 dB, speaker-to-mic replay). Because real and fake clips get the same treatment, the channel itself carries no label.

R2 · Partial fakes and localisation

- **Building 100 partial fakes:** take a real 6–10 s clip of a test speaker and replace one 1–3 s stretch with a fake clip of the same speaker (In-the-Wild has both for every person). Loudness is matched

and the joins get a 20 ms crossfade so there is no click to detect. The exact fake interval is saved as ground truth.

- **A segment head:** a second small classifier trained on random 1 s crops (clean and channel-augmented) from training speakers only.
- **Localisation:** slide a 1 s window every 0.5 s. Each 0.5 s cell gets the mean score of the windows covering it. A cell above the threshold is marked synthetic. The threshold is chosen on 60 separate partial fakes from calibration speakers, never on the test set.
- **Metric:** precision, recall and F1 over 0.5 s cells across all 100 test partial fakes.

R3 · Streaming mode

- The browser captures microphone audio as raw 16 kHz samples and sends 500 ms chunks over a WebSocket.
- The server keeps a rolling buffer for that call and, every 500 ms, scores the latest 2 s of speech (EER at 2 s was 2.56% in Phase 1).
- The UI shows a live score line, the current verdict and how long the voice has looked synthetic. A smoothed score avoids flicker.
- A "Simulate call" button streams a demo file in real time, so the demo works even without a microphone.

```
browser mic ──500 ms PCM chunks──► WS /ws/stream
                server: rolling 2 s buffer → XLS-R → head → score
◄── {t, score, smoothed, verdict} every 500 ms
```

R4 · Phase 1 guard

After every change the Phase 1 experiments are re-run on the same test speakers. The new model is accepted only if each Phase 1 EER is within 2 percentage points of the Phase 1 model.

3. Results (filled automatically)

R1 · Channel robustness

Condition	EER	× clean
Clean	1.80%	1.00×
8 kHz resample	1.40%	0.78×
AMR-NB 12.2k	1.80%	1.00×
Opus 12k	2.10%	1.17×
Noise 10 dB	3.50%	1.94×
Random mix	11.11%	6.17×

Worst case: 1.94× clean, target ≤ 2.5× → PASS

R2 · Localisation

100 partial fakes, 0.5 s cells: precision 79.95%, recall 81.02%, F1 80.48% (threshold 0.560 from calibration fakes).

R3 · Streaming

Update interval 500 ms; median processing per update 25.7 ms, p95 31.4 ms (must stay below 500 ms).

R4 · Phase 1 guard

Phase 1 test	Phase 1 model	Phase 2 model	Change
Unseen speakers 4 s	2.31%	2.01%	-0.30 pts
2 s clips	2.56%	2.26%	-0.30 pts
10 s clips	2.26%	2.01%	-0.25 pts
Phone line	2.85%	2.96%	+0.10 pts

4. Build status

Step	Status
Codec, 8 kHz, noise and channel-mix functions (real AMR-NB / Opus via ffmpeg)	Done, tested
Feature extraction: 2,000 test clips × 6 conditions	Done
Feature extraction: channel-augmented training set + 1 s segment crops	Done
Re-train detector with channel augmentation + modern TTS + clean real speech	Done
100 + 60 partial fakes, segment head, F1	Done
Streaming WebSocket + live UI + simulate-call	Done
Tests for all of the above, deck + document update, final zip	Last